CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

PROPOSAL PRESENTATION

Planner Copilot

A local AI assistant for production planning simulations

Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas

Team 1 · Guided by Dr. Hamidreza Heidari · 20 October 2026

THE PROBLEM

Good production planning needs simulation, and experts

A busy week, missing workers or a broken machine: a simulation shows what will happen before it does.

Today mostly only a simulation expert can run it correctly, and small companies often have none.

An AI chatbot alone cannot plan production, and nobody can check its answer without an expert.

Key motivations

- Small companies get answers without a simulation expert
- Large companies save expert time
- Company data stays in-house
- Reliable AI help for planning is needed now

TECHNISCHE HOCHSCHULE DEGGENDORF

PROJECT SCOPE

What we will and will not do

Aim

Test whether a local AI assistant with code checks can answer planning questions at the level of published expert simulations.

Includes

- About 4 published simulation cases
- Test questions with correct answers
- A local AI assistant with code checks
- Comparison of three setups

Out of scope (current phase)

- A company’s own production data and models
- Commercial simulation tools
- Use in a real factory

GOALS

Three measurable targets

≥ 90%

Close to the expert

For questions with an answer, the AI’s answer is close to the expert’s answer.

≤ 2%

Confidently wrong

The AI gives a wrong answer and sounds sure. This is the most dangerous error.

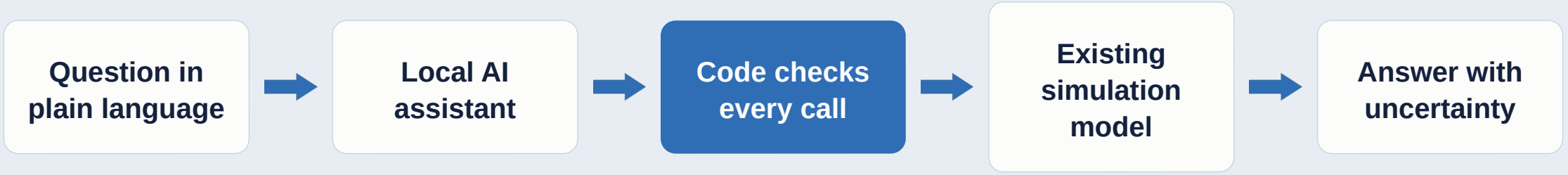
≥ 90%

Correct refusals

When a question cannot be answered, the AI says so instead of guessing.

OUR IDEA

An AI assistant that runs the simulation, and checks itself



Example: “Busy week, +20% orders. Do we need a fourth worker?”

TECHNOLOGY STACK

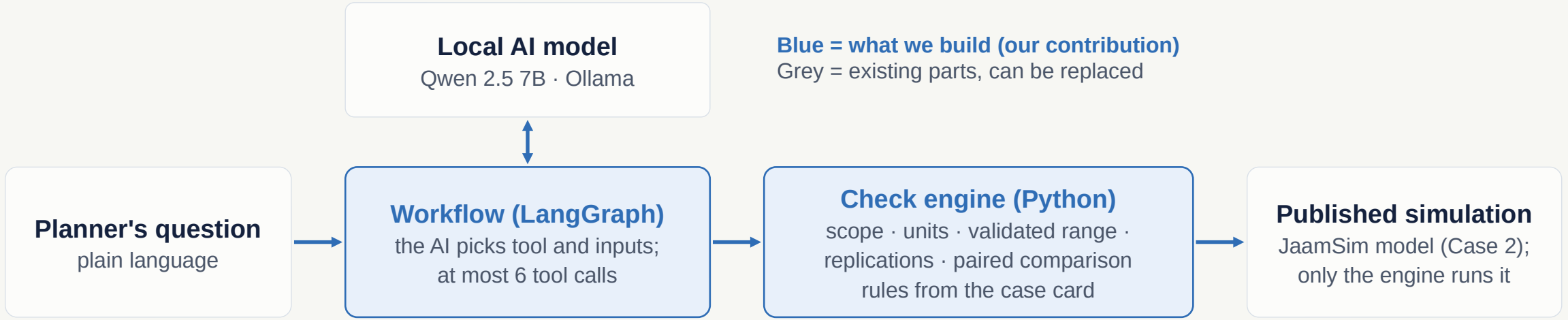
Process	Tool	Purpose
Simulation	JaamSim, SimPy, Python	Factory models, each in the tool its authors published it in
AI model	Open-source LLMs (Qwen, Llama, Gemma) via Ollama	Understands plain-language questions; runs on our own computer
Workflow	LangGraph	Fixed order of steps: the AI picks a tool and inputs, code checks and runs it
Code checks	Python (NumPy, SciPy), MCP server	Five checks on every simulation call; refuses out-of-scope questions
Interface	Gradio (planned)	A simple web page to ask questions and see the answers
Compute	Free cloud GPUs (Colab, Kaggle)	A computer with a graphics card to run the AI
Code sharing	GitHub	Code history, issues board, progress for the lecturer

Tools may change as the project progresses !

TECHNISCHE HOCHSCHULE DEGGENDORF

MVP · WHAT ALREADY WORKS

Three real answers from our first version



① Normal question → answer
"Normal week: hours after the Friday shift?"
→ 3.37 h · expert result 3.31 h ✓

② Outside validated range → warning
"+20 % orders, only 2 workers?" → ≈ 38 h
⚠️ outside 3–4 workers, indicative only

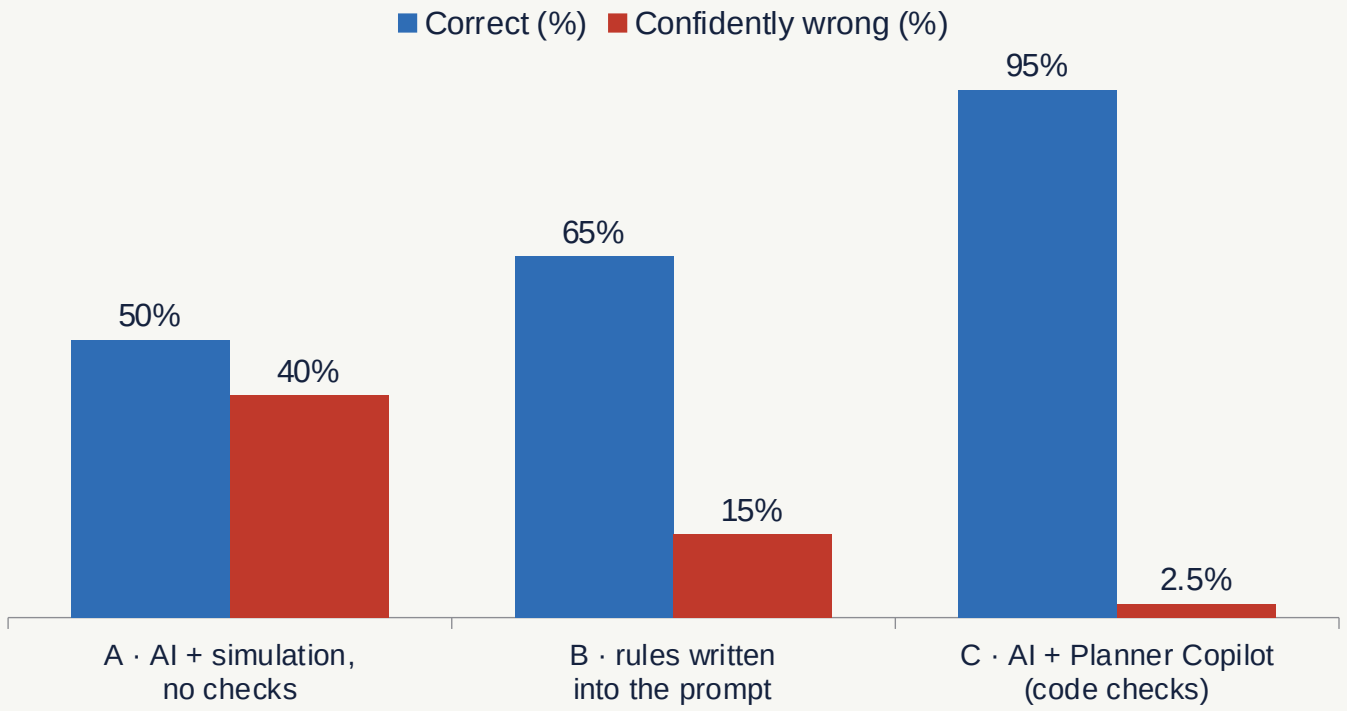
③ Outside the model → refusal
"Overtime cost in euros?"
→ refused: the model has no cost data

Case 2 (SME job shop) · development run, 17 Oct 2026 · every number is produced by the simulation; warnings are added by code

TECHNISCHE HOCHSCHULE DEGGENDORF

MVP · FIRST FEASIBILITY TEST

Code checks make the difference



Confidently wrong answers: 16 → 1 out of 40.

Rules in the prompt help (16 → 6); checks in code help much more (16 → 1).

Small test: 40 questions, 2 development cases, Qwen 2.5 7B via Ollama on a free cloud GPU (16 Oct 2026).

Full experiments: about 150 questions, 4 cases, several models (P4, 8 Dec).

Same 40 questions for all three setups · each answer graded correct / cautious / confidently wrong · details: github.com/bodhi584/planner-copilot

TECHNISCHE HOCHSCHULE DEGGENDORF

PLAN

Four steps to get there

1 · Find cases

About 4 published simulation cases from different industries, using public data only.

2 · Rebuild and ask

Rebuild each case and write test questions with the correct answers.

3 · Build the assistant

A local AI assistant with code checks, built with open-source tools.

4 · Compare

Expert result as reference. Three setups: AI + simulation, + rules in the prompt, + Planner Copilot.

TEAM

Who does what

Yu Han

Team lead · core code

Case 2
Core code: case cards, checks, workflow
MVP, feasibility test
Tests, CI, packaging
Coordination

Melisa Subasi

Questions & report

Case 1
Test questions
Final report editor

Meron Hagos

Interface & literature

Case 3
User interface
Literature review

Alejandro Rivas

Experiments

Case 4
Experiments
GitHub & demo

A case is one published factory simulation model with known expert results.

TIMELINE

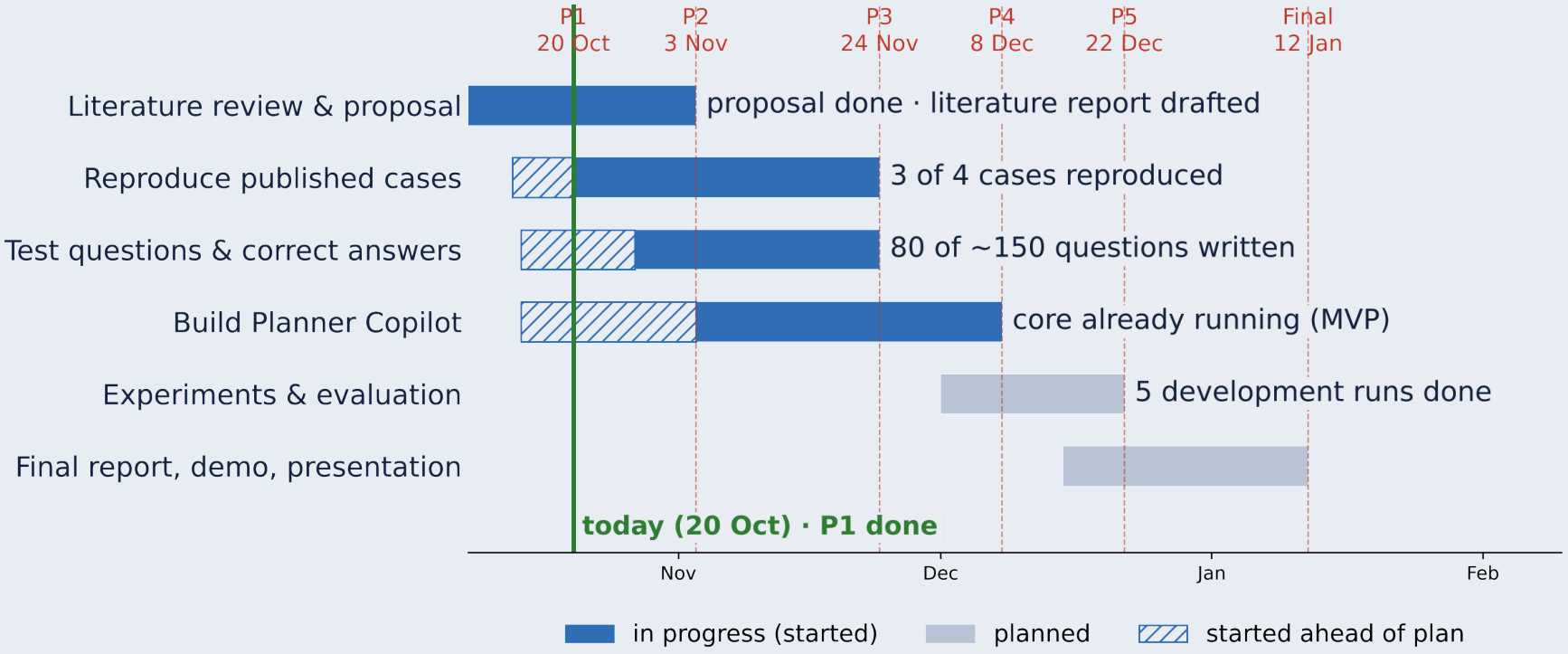
From proposal to final hand-in

Date	Milestone	Status
20 Oct	Proposal, team, GitHub, MVP	✔ **done**
3 Nov	Literature report, core code, baseline simulation, test plan	in progress: 3 of 4 cases reproduced, literature report drafted, core code running
24 Nov	Cases rebuilt, first version of the assistant	planned
8 Dec	First experiments	planned
22 Dec	Demo, experiments and analysis, report draft	planned
12 Jan	Final presentation, report, demo video	planned

TECHNISCHE HOCHSCHULE DEGGENDORF

GANTT CHART

Project schedule



EXPECTED OUTCOMES

What we will hand in

Test questions

Questions with correct answers for about 4 published cases.

Planner Copilot

A working open-source assistant that runs the simulations.

Comparison

Measured results for three setups against our targets.

Report and demo

Final report, demo video and a GitHub repository.

BILL OF MATERIALS

What the project needs

Component	Specification	Cost
GPU access (Colab, Kaggle)	Cloud GPU for the local AI	Free tier
Simulation cases and data	Published, public	Free
Software	Open source	Free

RISKS, ETHICS

Keeping it safe and realistic

Ethics

Only public data, so no privacy issue. All sources are cited.

Main risk

Rebuilding the cases may take longer than planned. Fallback: fewer cases.

Thank You !

Team 1 · Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas

github.com/bodhi584/planner-copilot